

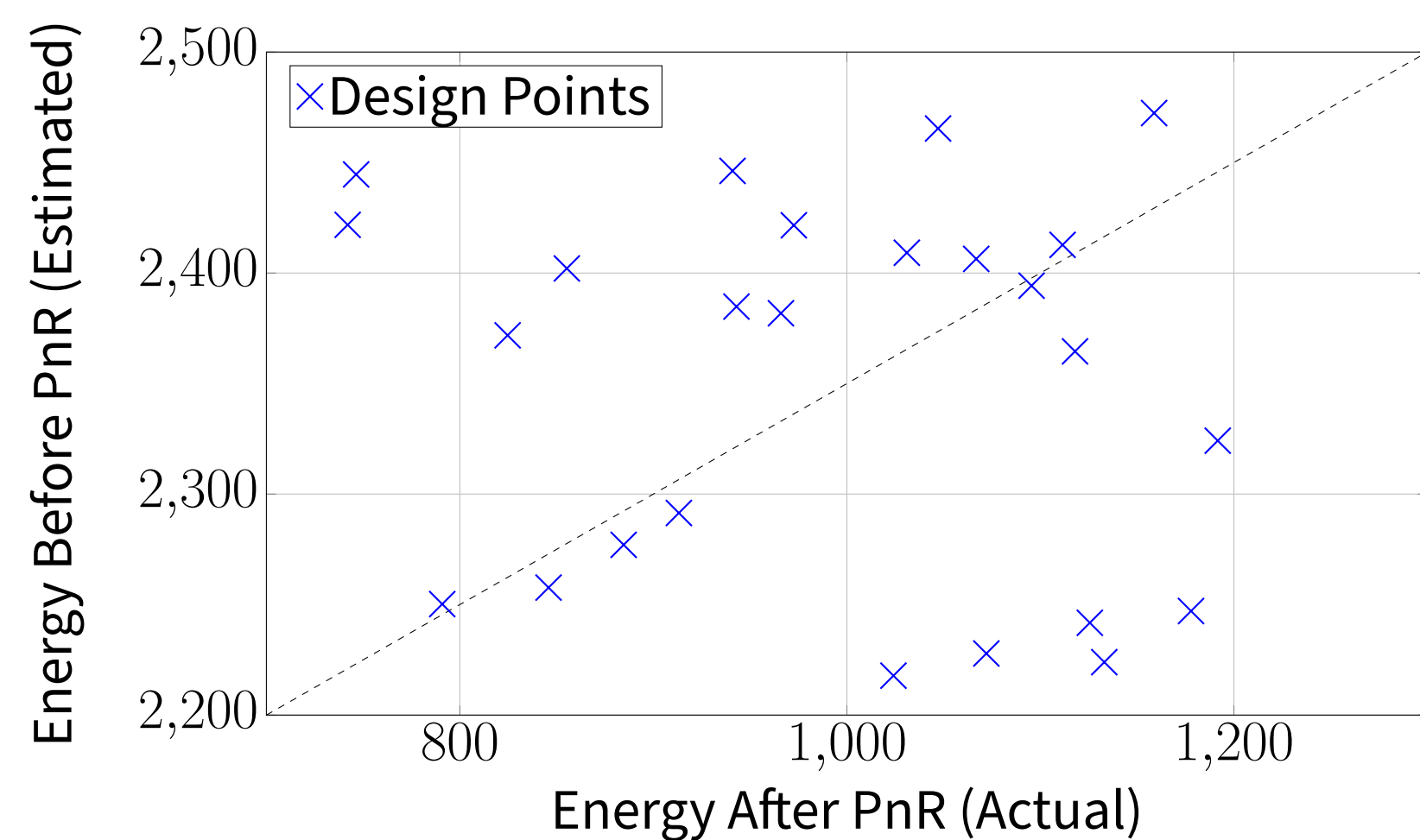
AN AGILE AND ACCURATE ENERGY ESTIMATION METHODOLOGY FOR CGRA-MAPPED ALGORITHMS: THE SYNCHOROS APPROACH

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1. Problem Statement

Enable CGRA energy estimation that is:

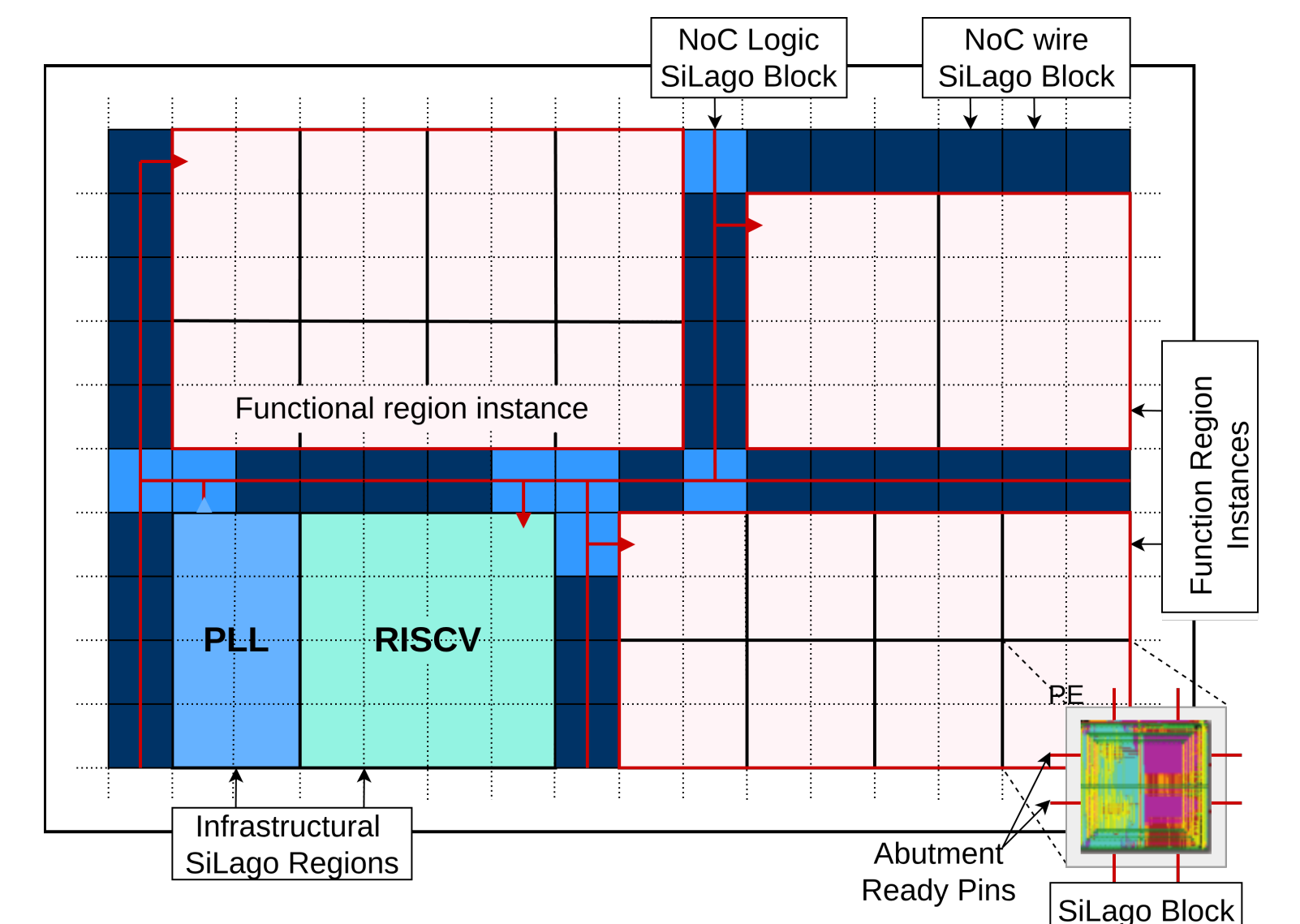
- *post-route-accurate*,
- *consistent* across functionality, dimensions, and parallelism



2. Our Solution

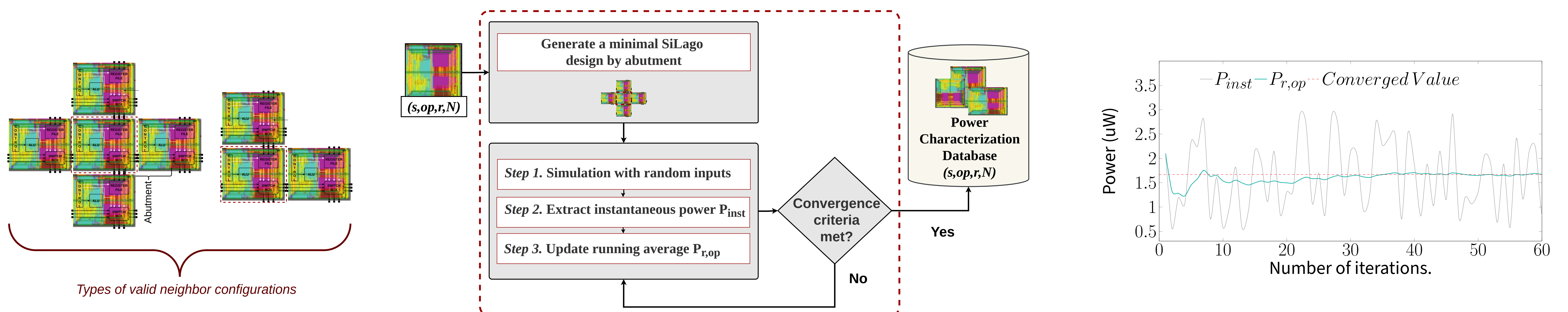
Synchoros ($\chi\rho\rho\acute{o}\varsigma$) characterization.

CGRA functional regions composed by abutment of SiLago blocks at the GDSII level on a virtual grid.



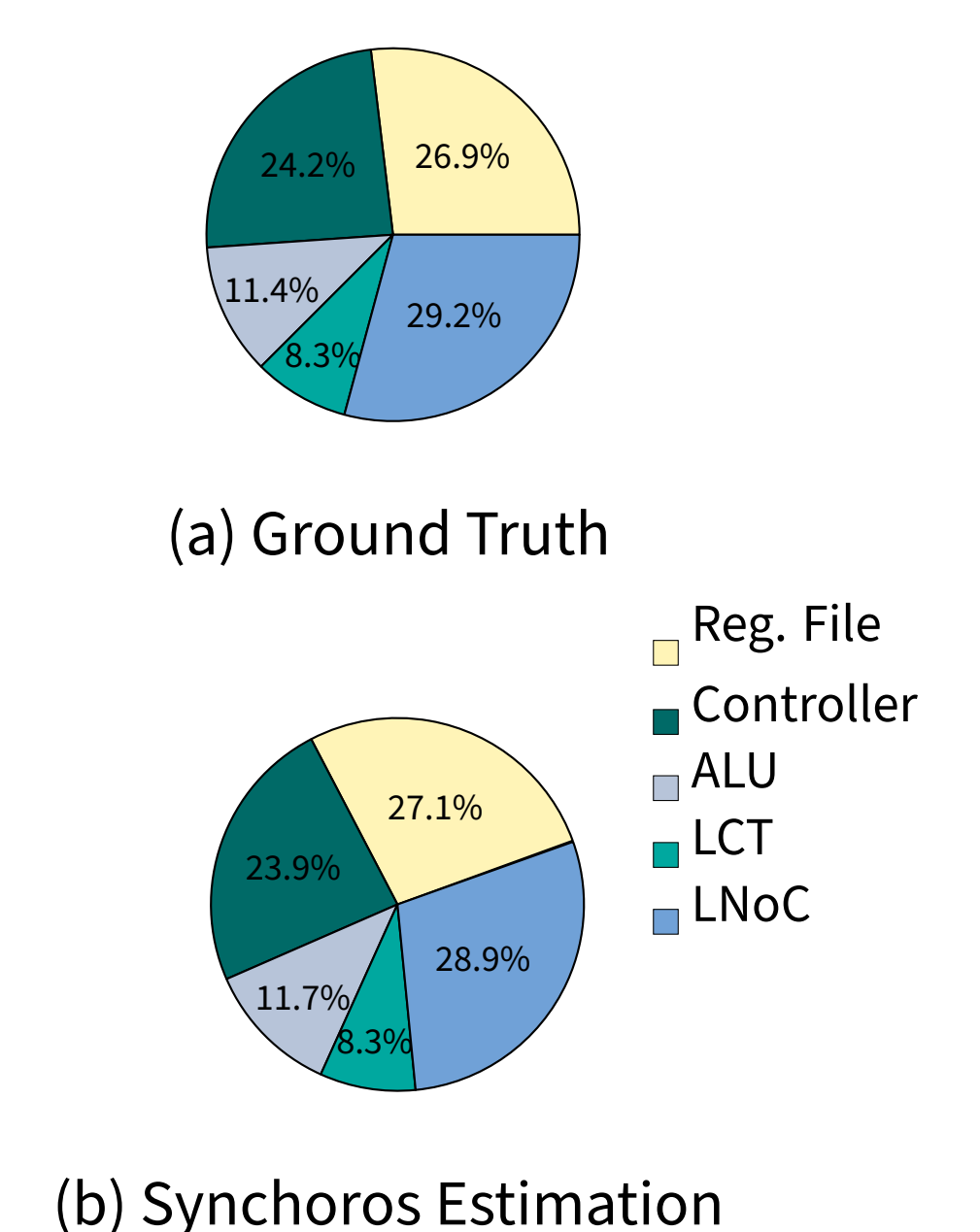
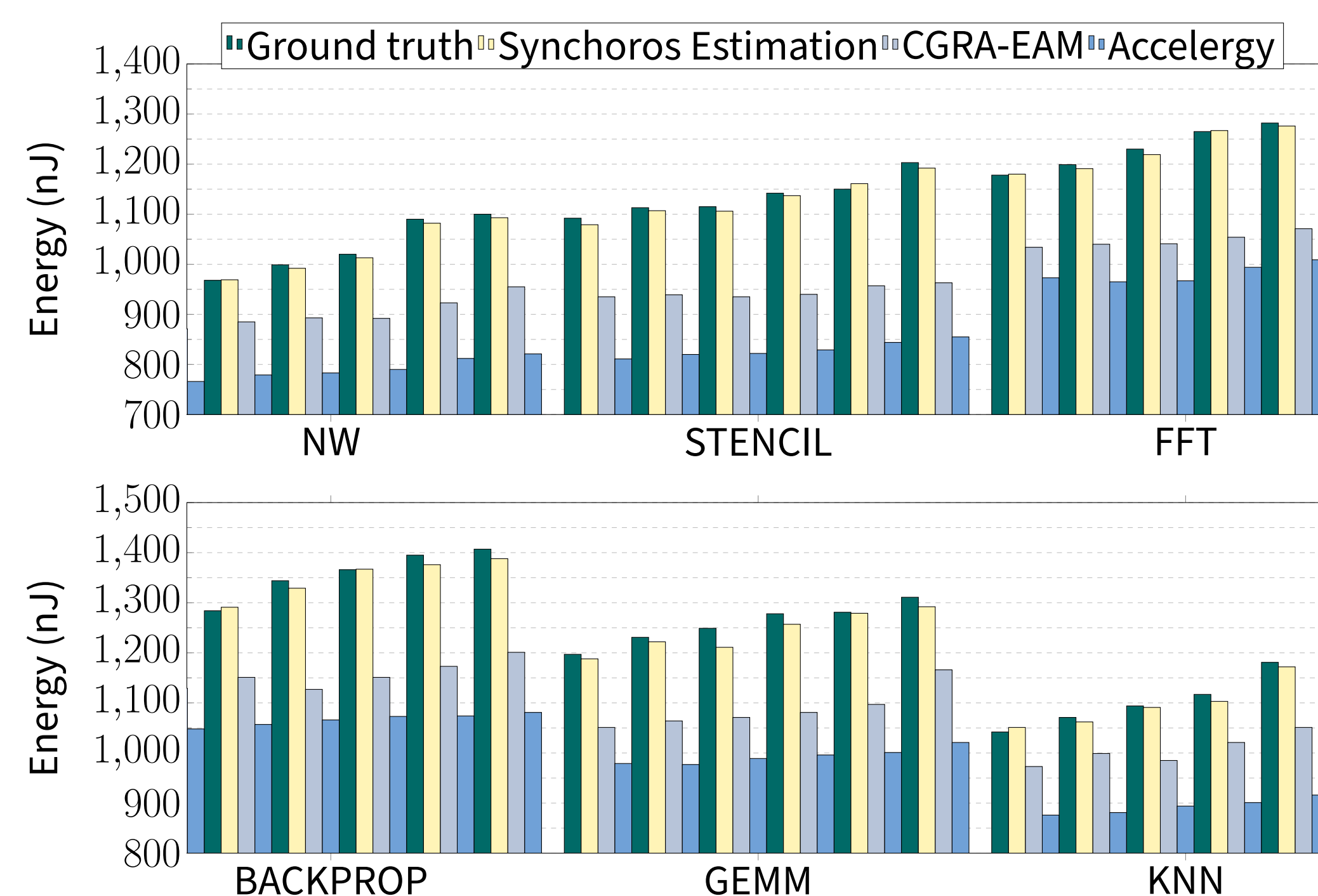
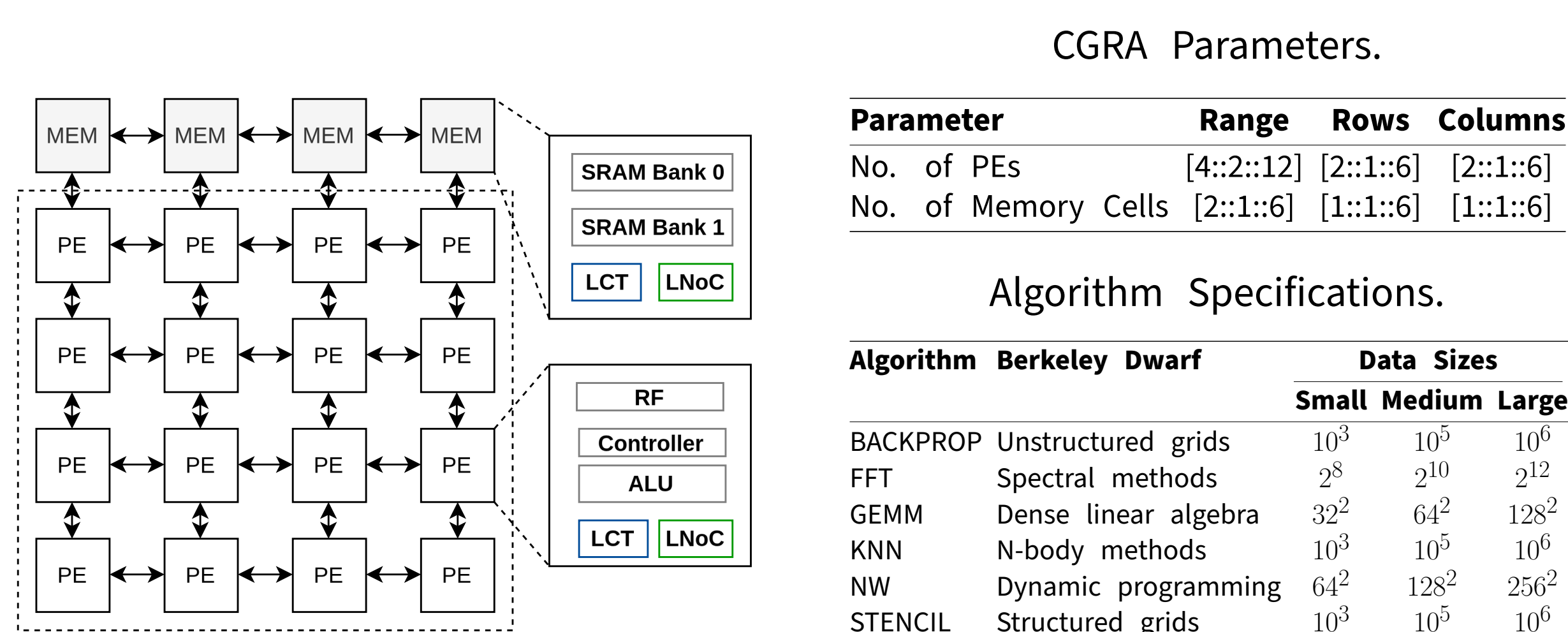
3. Synchoros Characterization Methodology

Energy characterization with spatial context information from valid neighbors for accurate parasitics extraction.



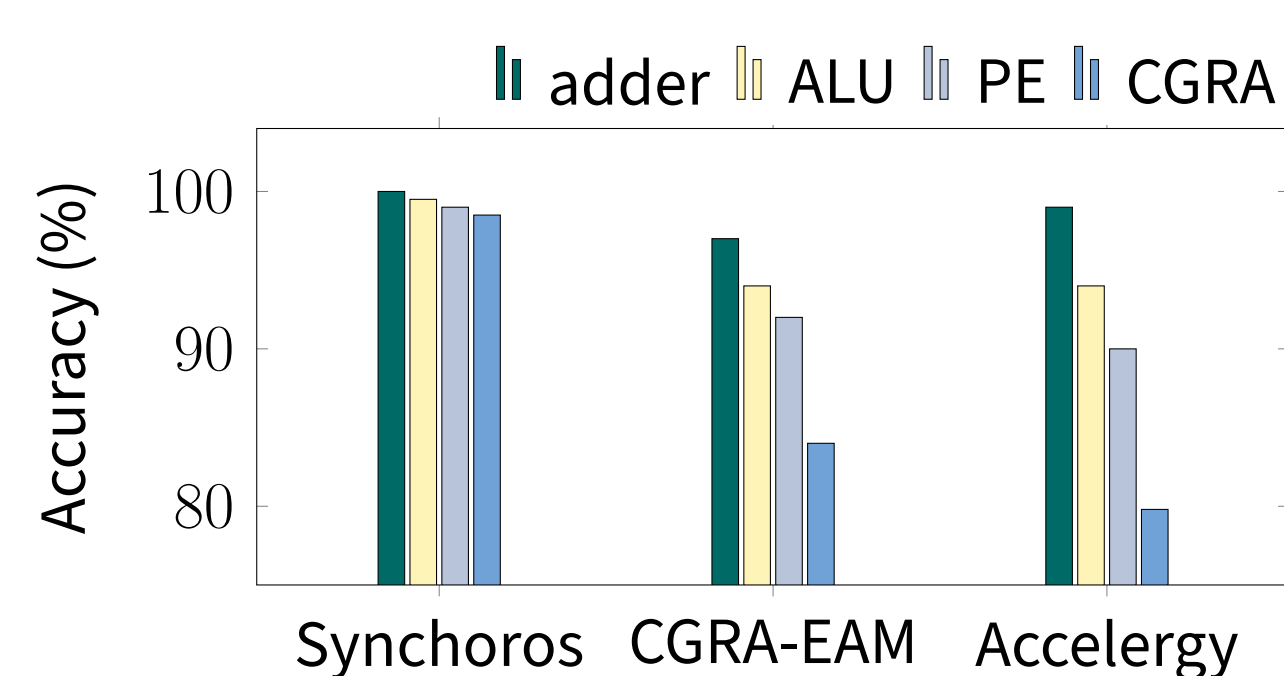
4. Experimental Results

Estimation accuracy of 98% relative to post-route results on DRRA + DiMarch CGRA.



5. Scalability and Overheads

- *Consistent* accuracy at different hierarchical granularities.
- *Modest overheads* for abutment requirements in three state-of-the-art CGRAs.



Overheads of Synchoricity

Design	Std. cell area (μm^2)	Energy (nJ)
(4x4 PEs)	asynchoros	Synchoros
DRRA+DiMarch	92512.6	97786.3
HycUBE	87376.4	94055.2
ADRES	84821.1	90286.1

6. Conclusion

- Synchoros energy estimation is *post-route-accurate*, explicitly accounting all wires in a CGRA.
- *Consistent* across functionality, dimensions, parallelism, and hierarchical granularities.
- Follow-up work planned to extend synchoros estimation to the application level for an SoC.

